

Decision Tree Classification

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In [45]: import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

```
In [46]: data = pd.read_csv('drug.csv')
data.head()
```

```
Out[46]:
```

	Age	Sex	BP	Cholesterol	Na_to_K	Drug
0	23	F	HIGH	HIGH	25.355	DrugY
1	47	M	LOW	HIGH	13.093	drugC
2	47	M	LOW	HIGH	10.114	drugC
3	28	F	NORMAL	HIGH	7.798	drugX
4	61	F	LOW	HIGH	18.043	DrugY

```
In [47]: data.isnull().sum()
```

```
Out[47]: Age          0
Sex            0
BP            0
Cholesterol    0
Na_to_K        0
Drug          0
dtype: int64
```

No missing values so we can continue with splitting

```
In [48]: label_encoder = {}
for column in ['Sex', 'BP', 'Cholesterol', 'Drug']:
    le = LabelEncoder()
    data[column] = le.fit_transform(data[column])
    label_encoder[column] = le
```

```
In [49]: X = data.drop('Drug', axis=1)
y = data['Drug']
```

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In [50]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
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In [51]: clf = DecisionTreeClassifier(criterion='entropy', random_state=42)
clf.fit(X_train, y_train)
```

```
Out[51]: ▼ DecisionTreeClassifier ⓘ ?
DecisionTreeClassifier(criterion='entropy', random_state=42)
```

```
In [52]: y_pred = clf.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy * 100:.2f}%")
```

Accuracy: 100.00%

```
In [53]: plt.figure(figsize=(12, 8))
plot_tree(clf, feature_names=X.columns, class_names=label_encoder['Drug'].classes_, filled=True)
plt.show()
```

